

and the colon. Python's philosophy is different, which is, "There is only one way to do the things and it's called the Pythonic way." The current version of Python is 3.6.0.

Advantages.

- One of the best things about Python is that it is well documented.
- Python is used in 3D animation and game development, which differentiates it from other languages.
- In school, students learn Python quicker than C or C++ because it is easier.
- Python uses white space indentation, while others use curly braces. Comparatively, Python code is neater and cleaner.
- Code written in Python is very short compared to other languages, which demonstrates its power.
- Python also supports JVM so code written in Python can easily work with some of the Java objects/APIs.
- If you are new to programming, learn Python because by learning this one language, you can easily jump into any field of interest, whether it's automation, Web development, game development or data analytics. Python is everywhere.

Disadvantages.

- Indentation is the biggest issue in Python. The code will not work and sometimes people get irritated because of it.
- Python forces programmers to follow a particular convention.
- It's slower than older languages like C and C++.
- It is also an interpreted language.
- It doesn't perform multi-processor/multi-core work efficiently.

Popularity. Python was bestowed the *TIOBE Programming Language of the Year* award in 2007 and 2010. The TIOBE index measures the growth and popularity of the language over a period of one year.

Popular frameworks. Django, TurboGears, Web2py, Bottle, CherryPy, Flask and Pyramid are some of the popular frameworks.

Syntax. Python syntax is very neat and clean:

```
#!/usr/bin/python
```

```
print "Hello! Welcome to the world of Python!"
```

The extension of the file is .py.

So which is the best programming language to learn and use?

There is no specific answer to this question because you can perform almost all tasks with any of the three languages. It depends on what you want to do and how you can do it efficiently.

If Web development is the first priority, then PHP is the first preference. For test automation and scripting, Python and Perl are very popular. There are lots of frameworks available in all three languages, and since community support is good, you will never be stuck with a particular problem.

PHP has lots of popular CMS frameworks available like WordPress, Joomla and Drupal. Any newbie can deploy websites easily using these frameworks. There are lots of plugins available for these frameworks and almost all the major functionalities are covered. Similarly, Python has the Django framework, which is very popular and most Python lovers use it for Web development.

My advice would be to learn all three languages and use them based on the situation. Beginners can start with Python as it is very easy, and later jump into learning PHP and Perl, in that order. The basic logic for all the programming languages is the same and the only difference is in terms of the syntax. Usually, the person who knows one language can easily migrate to another language very quickly.

There are never ending discussions on all the forums about which is the best programming language for career growth. My advice would be to learn one language very well. Pick the one you feel comfortable with and slowly venture into exploring the other two.

My preference

I will go with Python as it is evolving very fast and is easy to learn. The use cases of Python cover a very broad range. It is used in almost every IT segment and, by learning this language, you can do wonders in all those segments.

The most important thing is to learn the Pythonic way of writing a code. Here are a few examples:

Example 1. The 'non-Pythonic' way of writing a code is:

```
def test(a, b):
    a[0] = 2
    b[0] = 7.5
abc = [0]
xyz = [0]
test(abc, xyz)
abc = abc[0]
xyz = xyz[0]
```

The Pythonic way of writing the same code is:

```
def test():
    return 2, 7.5
alpha, beta = foo()
```

Example 2. The non-Pythonic way of writing code is:

```
i = 0
while i < mylistlen:
    do(mylist[i])
    i += 1
```

The Pythonic way of writing the same code is:

```
for i in range(mylistlen):
    do(mylist[i])
```

Another Pythonic way of writing the same code is:

```
for item in mylist:
    do(item)
```

Example 3. A short program for reversing a string is shown below:

```
'Hello Python'[::-1] EPY
```

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